Topics in Political Methodology

PLSC 504

Exercise One

August 31, 2026

Part I: Separation, Simulated

In our class discussion on Monday, we discussed what happens when we encounter "separation" / "perfect prediction" in a logistic regression context. Here, we're going to explore that further using simulations, and in particular focus on the differences between standard and Firth / penalized logistic regression in the presence of total and near-separation.

1. Consider a large population of $\mathfrak{N}$ cases with two binary variables X and Y, where in the population there are equal numbers of individuals with $\{X = Y = 0\}$, $\{X = Y = 1\}$, and $\{X = 1, Y = 0\}$ but *no* individuals that have $\{X = 0, Y = 1\}$:

	$Y = 0$	$Y = 1$
$X = 0$	$\mathfrak{N}/3$	0
$X = 1$	$\mathfrak{N}/3$	$\mathfrak{N}/3$

   Begin with a small (say, $N = 20$) sample from this population. Fit standard logistic and Firth-corrected logit models of Y on X to the sample, and recover estimates of the estimated logit slope ($\hat{\beta}_X$) and its associated standard error for each. Describe the distribution of the estimates of each,[1] and compare them across the two models.

2. Increase the sample size N by a factor of 10, and repeat part (1), discussing if and how a larger sample changes the resulting estimates from both types of models.

3. Increase the sample size N further by a factor of 20, and repeat part (2).

4. Repeat parts (1)-(3), but now sampling from a population where one in 100 individuals have $\{X = 0, Y = 1\}$:

	$Y = 0$	$Y = 1$
$X = 0$	$\mathfrak{N}/3.030303$	$\mathfrak{N}/100$
$X = 1$	$\mathfrak{N}/3.030303$	$\mathfrak{N}/3.030303$

   In particular, discuss how the standard and Firth logit models differ in this case, and how those differences change as the sizes of the samples increase.

[1] Note: A single simulation doesn't really tell you anything. What I'm asking for here is basically to $\{$sample $\rightarrow$ fit the model to the sampled data $\rightarrow$ extract the slopes and their standard errors$\}$ $\leftarrow$ do that some large number of times, and then summarize – via plots and/or tables – the values of those quantities across the many samples / simulations.

Part II: "A covert ethnic-pride celebration for red-state whites of Northern European descent."

In a 2019 publicity stunt, a Fox News commentator began her show by attempting to drink a light-bulb-bedazzled steak through a straw. In her commentary on the bit, she noted that "it (the steak) has everything the Democrats hate. If I could have put an SUV on this, I would have."

What does this have to do with political science? The answer ought to be obvious.

Waaay back in 2005, ABC News and the Washington Post commissioned a poll about public opinion on traffic. Among other things, pollsters asked 1204 lucky, randomly-selected Americans:

> "What kind of vehicle do you usually drive – a car, an SUV, a pickup truck, or what?"

In this part, we'll explore the political dynamics of car ownership, using the data from the 2005 ABC/WP poll. The main variable of interest is `cartype`, coded one for cars, two for SUVs, and three for pickup trucks. Covariates include dummy variables for `urban` residence, being `married`, having `kids`, and being `black` and/or `female`, as well as a naturally coded variable for `age` and an ordinal variable for level of `education`. Best of all, we also have two dichotomous variables for political party (`democrat` and `GOP`, with independents as our baseline) and a four-point ordinal scale indicating each respondent's approval or disapproval for then-President Bush (II).

Exercise

1. After providing some summary statistics, estimate a multinomial logit (MNL) model of vehicle type ownership. Report your estimation results, and interpret these findings, in statistical and substantive terms. Are the results in the "expected directions"? Discuss their statistical significance.

2. Using the MNL results, generate and examine the predicted probabilities of each type of vehicle, across the range of values for *one* of your more important independent variables, using tables or graphs of the probabilities. Interpret these results in substantive terms.

This assignment is due *electronically*, as a *PDF file*, at 11:59 p.m. EST on Friday, September 18, 2026. In addition to your responses, please include a copy of all computer code used to generate summary statistics, models, plots, etc. This can be in any form: a separate `.R` file, an appendix in the PDF, or as a `.Rmd` or similar format containing both content and code. You can submit your homework by emailing copies **both** to Dr. Zorn (`zorn@psu.edu`) and Ms. Bandeira (`lfb5423@psu.edu`). This assignment is worth 50 possible points.